Resilience Academy

SDI Training for ICT Professionals

Adding Your Data to CRD

Upload, Describe, Share, and Connect

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Day 2 | 30 Minutes (Lecture) | crd.resilienceacademy.ac.tz

Resilience Academy

Session Overview: Adding Your Data to CRD

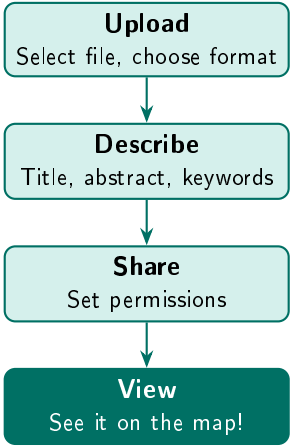
Your CRD is installed and customised. Now it is time to fill it with data!

What we will cover:

1. How to upload data via the web interface
2. Which file formats are supported
3. Setting metadata: title, description, keywords
4. Controlling who can see and edit your data
5. Viewing uploaded layers on the map
6. Connecting QGIS to your CRD (brief intro)

The upload process has 3 stages:

1. **Upload** the file
2. **Describe** it with metadata



How to Upload: Step by Step

Starting the upload:

1. Log in to your CRD
2. Click **"Data"** in the top menu
3. Click **"Upload Layer"** (blue button, top-right)
4. You will see a file upload form

Uploading a Shapefile:

5. Click **"Choose Files"**
6. Select **all parts** of your shapefile:
   - `.shp` — the shapes (required)
   - `.dbf` — the attribute table (required)
   - `.shx` — the shape index (required)
   - `.prj` — the coordinate system (required)

Uploading a GeoTIFF (raster image):

1. Same steps, but select a single `.tif` file
2. CRD recognises it automatically as a raster layer
3. Processing may take longer for large files

Uploading GeoJSON:

1. Select a single `.geojson` or `.json` file
2. GeoJSON files are simpler — only one file needed
3. Great for web-generated data

Common Mistake

Supported Formats: What Can You Upload?

Vector Formats (points, lines, polygons):

Shapefile (.shp + .dbf + .shx + .prj)
Most common GIS format, used by everyone

GeoJSON (.geojson)
Single text file, web-friendly, easy to inspect

KML (.kml)
Google Earth format

CSV (.csv with lat/lon columns)
Spreadsheet data with coordinates

Raster Formats (grid/image data):

GeoTIFF (.tif / .tiff)
Standard for satellite images, elevation, rainfall

ASCII Grid (.asc)
Simple text-based raster format

Analogy: Raster data is like a **photograph** — it is made of tiny squares (pixels), and each pixel has a value (e.g. elevation in metres, rainfall in mm).

Which format should I use?

- **Buildings, roads, boundaries? →**

Setting Metadata: Make Your Data Findable

After uploading, CRD asks you to **describe your data**. This is called **metadata** — "data about your data."

Think of metadata like a book's catalogue card:

- A book without a title and author is useless on a library shelf
- Data without metadata is useless in a catalogue

Required fields:

- **Title**: Clear, descriptive name
  Good: "Flood Hazard Zones — Tanga City 2024"
  Bad: "data_final_v2"

Recommended fields:

- **Keywords**: Searchable tags
  e.g. flood, hazard, Tanga, 2024, urban
- **Bounding box**: Geographic extent
  Usually auto-detected from the data
- **Attribution**: Who created the data?
  e.g. "Tanga City Council, Planning Dept."
- **Date**: When was the data collected or published?
- **License**: How can others use it?
  e.g. Creative Commons, Open Data Commons

Permissions: Who Can See and Edit Your Data?

CRD lets you control access at **three levels**:

Public (Anyone)

Everyone can view and download, even without logging in. Use for open data that should be widely available.

Group (Specific Teams)

Only members of selected groups can view or edit. Use for department-level data sharing.

Private (Only You)

Only you and admins can see it. Use for draft data or sensitive information.

How to set permissions:

1. After uploading, go to the dataset page
2. Click **"Change Permissions"** (or the lock icon)
3. Choose who can:
   - **View**: see the data and download it
   - **Download**: access the raw files
   - **Edit**: change metadata and style
   - **Manage**: change permissions and delete
4. Add users or groups as needed
5. Click **Save**

Practical Example

Scenario: You upload flood data for Tanga

Viewing Your Uploaded Layers on the Map

Once uploaded, your data **automatically appears on the map viewer**.

What you can do:

- **Preview**: Click the layer name to see it on the interactive map
- **Zoom**: The map auto-zooms to your data's extent
- **Identify**: Click on any feature to see its attributes in a popup
- **Style**: Change colours, labels, and symbols through the web interface
- **Compose**: Combine multiple layers into a saved map

Layer styling — change how data looks:

Colours: Choose fill and border colours
e.g. red for high risk, green for low risk

Labels: Show names or values on the map
e.g. ward names, building types

Symbols: Change point shapes and sizes
e.g. circles for schools, squares for hospitals

Classification: Group values

Connecting QGIS to Your CRD: WMS and WFS

Why connect QGIS to CRD?

- View CRD layers **directly in QGIS** — no downloading needed
- Always see the **latest version** of the data
- Combine CRD data with your **local files** for analysis
- Works over the network — even from another office

Adding a WMS layer in QGIS:

1. Open QGIS
2. Menu: **Layer** → **Add Layer** → **Add WMS/WMTS Layer**

Adding a WFS layer in QGIS:

1. Menu: **Layer** → **Add Layer** → **Add WFS Layer**
2. Click **"New"** to create a connection
3. Name: `My CRD (WFS)`
4. URL: `http://localhost/geoserver/ows`
5. Click **Connect** → select a layer → click **Add**

WFS gives you the actual data — you can query, select, and export features.

WMS vs WFS Reminder

Summary: Day 2 Achievements

Session 4 Recap:

- ✓ Upload layers via **Data → Upload Layer**
- ✓ Supported formats: **Shapefile, GeoJSON, GeoTIFF**
- ✓ Good metadata makes data **findable and usable**
- ✓ Permissions control **who sees and edits** your data
- ✓ Layers appear **automatically on the map**
- ✓ QGIS connects to CRD via **WMS and WFS**

Full Day 2 Achievements:

You now have:

- A running CRD instance
- Branded with your logo
- Users and groups created
- Map centred on your city
- Data uploaded and shared

Looking Ahead to Day 3

Tomorrow we explore **metadata standards**